

Robust metamodel-based simulation-optimization approaches for designing hybrid renewable energy systems

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ABSTRACT

We propose two robust metamodel-based approaches for designing stand-alone hybrid renewable energy systems that combine conventional and renewable energy generators with battery storage. Since the cost and reliability functions of the stochastic system do not have closed-form expressions, they can only be estimated via simulation. In the *response surface methodology*, the simulation input/output data is used first to locally fit first-degree surrogate functions for each supply scenario to perform a stochastic gradient descent, before a second-degree function is fitted and optimized when the gradient becomes small. In contrast, the *global response surface technique* globally fits a convex quadratic surrogate function on the simulation input/output data, then uses its minimizer to iteratively refine the search space. To account for estimation errors, a distributional ambiguity set based on a linear ϕ -divergence function is constructed around the nominal probability distribution of supply scenarios, and the expected reliability-adjusted cost of the system is computed using the worst-case distribution in this set. The validity of the proposed approaches and the value of considering distributional ambiguity are demonstrated through a realistic case study of a remote community in Northern Ontario.

1. Introduction

With the steady growth in global energy demand and the rising concerns over environmental change, pollution, and depletion of fossil energy resources, renewable energy sources (RES) like solar and wind are posed to play a key role in our transition to a sustainable economy [1,2]. Stand-alone hybrid renewable energy systems (HRES) that combine locally available RES with conventional energy generation and/or energy storage technologies constitute a viable alternative for serving the energy requirements of remote and isolated communities when their connection to centralized power grids is not technically and/or economically feasible [3]. However, given the intermittent nature of RES and the high cost of large-scale energy storage, it is crucial that the design of HRES is optimized so they can deliver energy economically and reliably. Therefore, the problem of optimal design of HRES has received a considerable attention from the scientific community in the last two decades [4–8].

Simulation–optimization (SO) is a widely-used method for HRES planning [9]. SO is particularly suitable when the problem's objective and/or constraint functions are difficult to express explicitly (analytically), but can be evaluated in a black-box model through simulation for a particular input [10]. Generally speaking, SO approaches can be categorized into model-based and metamodel-based methods [11].

Model-based SO encompasses ranking and selection, gradient search methods (i.e., stochastic optimization), direct (derivative-free) search methods and random/heuristic search methods like genetic algorithms (GA), simulated annealing (SA) and tabu search (TS) [10]. On the other hand, metamodel-based or surrogate models approximate the unknown input/output relationship of the system using an easy-to-optimize function [12]. Response surface, Kriging and neural network methods are among the most widely-used metamodels [13].

Given the stochastic, nonlinear and dynamic nature of HRES operations, metamodel-based SO represents a promising framework for planning them [14]. However, the literature on HRES design using metamodels in general, and response surface methods (RSM) in particular, is still sparse. Ekren and Ekren [15] used RSM to optimally size an autonomous solar/wind HRES with battery storage by minimizing the system cost and loss of load probability (LLP). Subsequently, Ekren et al. [16] conducted a break-even analysis and compared a grid-connected system to a stand-alone system in terms of their net present values using a similar RSM. Ren et al. [17] presented a simulation-based strategy for optimal design of a renewable cooling and heating system, and devised a solution approach that integrates an integer-based GA with RSM.

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A key limitation of SO methods in the context of stochastic systems is that they require the probability distribution (or the underlying data-generating process) of the simulation inputs to be known with certainty, otherwise the solutions obtained might be sub-optimal [18]. In this sense, they are quite similar to sample average approximation (SAA) methods that are known to have poor out-of-sample performance, especially when based on small data samples [19], which is often the case in HRES planning. To reduce the sensitivity of SO models to the simulation parameters and to hedge against uncertainty about them, a robust approach was investigated by Dellino et al. [18,20]. Their proposed approach adopts Taguchi's view of the uncertain world, but replaces its statistical techniques to tackle robustness using metamodels by Kriging and RSM.

A less-conservative approach for dealing with parameter uncertainty that has gained significant interest recently is distributionally-robust optimization (DRO) [21]. In DRO, the expected value of a probabilistic cost function is optimized, where the expectation is taken with respect to the worst-case probability distribution from a distributional ambiguity set (DAS) that is constructed based on historical data or parametric information about the "true" distribution. Although DRO has found application in some energy optimization studies [22–24], DRO models used so far are based on explicit formulations that require the objective and constraint functions to be fully known, which is not the case in HRES design problems.

Although SO is the most commonly-applied approach to design HRES, the vast majority of SO models use metaheuristics that often fall into local optima and require excessive numbers of simulation evaluations [25,26]. Metamodel-based approaches are shown to outperform metaheuristics, especially for complex problems [27]. However, among the handful of studies that apply metamodel-based SO in the context of HRES design, e.g., [15,17], the issue of distributional ambiguity, arising when few supply and demand scenarios extracted from limited data are used in the simulation, is largely ignored, leading to HRES designs that perform sub-optimally in reality. On the other hand, classical DRO models that can alleviate this issue are based on canonical-form optimization (i.e., explicit objective function and constraints) and are not suited for SO methods. The current paper fills this research gap by showing how metamodel-based SO methods can be embedded in a DRO framework to provide robust HRES designs that significantly outperform designs based on classical methods that directly use the empirical probability distribution of scenarios. In particular, the contribution of this paper is twofold. First, it provides novel and generic design approaches that combine RSM and DRO to overcome the limitations of existing methods. Second, it demonstrates the superiority of the robust designs obtained using the proposed approaches through extensive numerical analysis.

We propose two novel approaches for robustly optimizing the design of HRES under ambiguity about the probability distribution of solar and wind energy supply scenarios. Since the objective function can be evaluated only through simulation, response surface metamodels are fitted on the input/output (I/O) simulation data using design of experiments (DoE) techniques. The first approach uses a classical RSM that locally approximates the response function and uses gradient information and a line search to update the best design, before performing and optimizing a second-order approximation in the last iteration. The second approach, in contrast, globally approximates the response function and optimizes the convex quadratic approximation by solving a DRO problem to refine the search space in each iteration until the algorithm converges to an optimal design. To model uncertainty in both cases, first, a manageable number of solar and wind supply scenarios are generated from raw data using k -means clustering, and their probabilities are estimated and used to generate an empirical distribution. Then, a ϕ -divergence ambiguity set is constructed around the empirical distribution of scenarios to account for estimation errors. The two proposed approaches are applied to a hypothetical (yet realistic) case study that entails designing a solar–wind–battery–diesel

HRES for an isolated community in Canada. Comparisons between the two approaches are conducted for both the nominal and the robust problems.

The remainder of this paper is organized as follows. The next section describes the HRES to be optimized and presents the mathematical formulation of its optimal design problem. Section 3 presents the k -means clustering method used to extract supply scenarios and the ϕ -divergence ambiguity set constructed based on their empirical distribution. Section 4 explains, in detail, the two approaches proposed to robustly designing a HRES: the *classical RSM* and the *global response surface technique* (GRST). The regression models used to fit the I/O data in each approach are presented in the same section, along with the way distributional ambiguity is incorporated into the design process. In Section 5, the numerical results obtained from applying the proposed approaches on a realistic case study are compared and thoroughly analyzed. Finally, conclusions and ideas for future research are provided in Section 6.

2. System description and problem formulation

This section provides a brief description of the system under consideration, including its operational rules. Then, it presents the notations used to formulate the HRES design problem, along with the mathematical formulation of the objective functions and constraints.

2.1. The HRES and its operational rules

The HRES of interest consists of photovoltaic panels (PV), wind turbines (WT), a battery bank (BB) and a diesel generator (DG). We assume that the types and characteristics of these system components are pre-selected by the planner based on the location of the system and the available technologies, whereas their numbers/capacities are to be determined by solving an optimization problem. Since the remote/isolated community to be served is not connected to the grid, it has to rely on available local resources to satisfy its energy demand. This is to be achieved while minimizing the annualized total cost of the system, which includes both the capacity cost of its components, amortized over their useful lives, and the expected annual cost of fuel consumed by the DG. Power outputs of the PV and WT are intermittent and unpredictable since they depend on meteorological variables (e.g., solar irradiance, wind speed, temperature), but the planner has access to some historical output data for both. On the other hand, and without loss of generality, we assume that the total demand for energy in each period (e.g., hour) of the year can be predicted with high accuracy. If this assumption does not hold, one can simply treat the stochastic demand in the same way PV and WT outputs are treated, as will be explained later. All generation and storage units, along with the load, are assumed to be connected to the same bus, i.e., power flow losses are ignored. This is a reasonable assumption for small-scale HRES typically used in remote communities.

HRES are usually operated according to preset rules that specify, for example, which energy source is to be utilized when there is a mismatch between RES supply and demand. The utilization priority of the system resources is illustrated in Fig. 1. Based on this priority, the operational rules of the system are described as follows: In any time interval, if demand exceeds the energy supplied by the renewable generators (PV & WT), the shortage is compensated by the BB within its discharge rate and energy limits. The remaining demand (if any) is met by the DG within its rated capacity, then any excess loads are curtailed. On the other hand, if the energy supplied by the renewable generators exceeds demand, the excess energy is used to charge the BB within its charge rate and energy limits, then any remaining energy is dumped. There rules ensure that the BB is never charged directly from the DG, that load curtailment happens only when all sources (WT, PV, BB & DG) are not enough to satisfy the demand, and that energy cannot be dumped if the demand is yet to be fully satisfied.

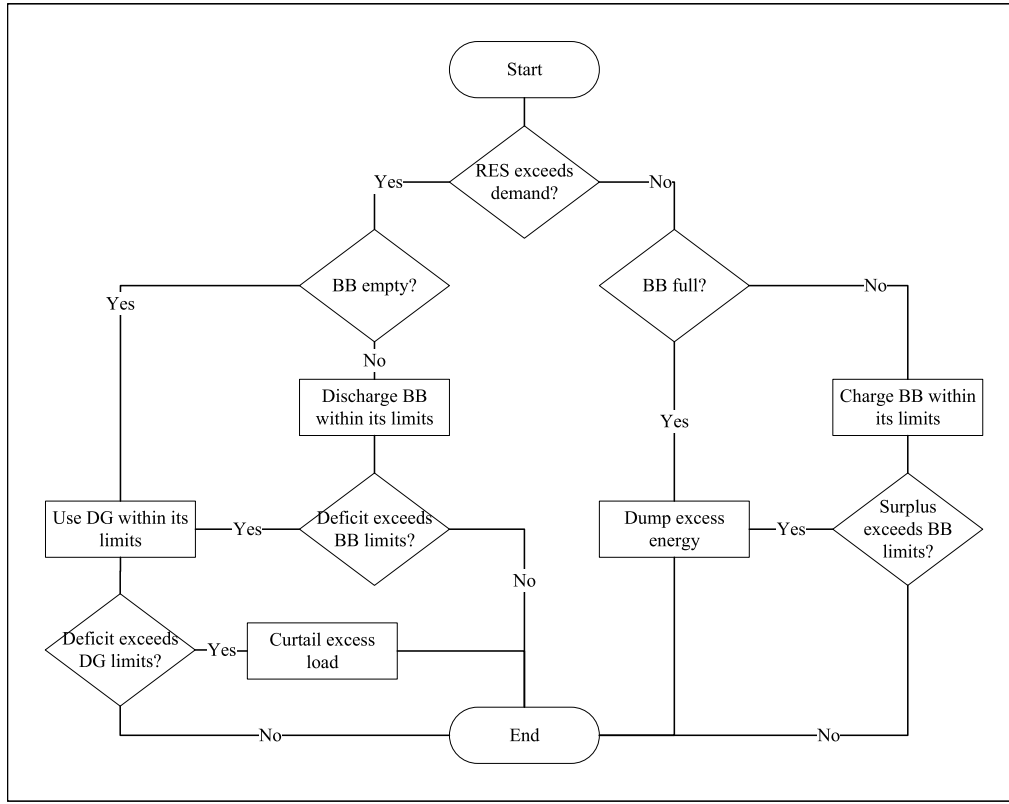


Fig. 1. Utilization priority of system resources.

2.2. Mathematical formulation of the HRES design problem

Table 1 depicts the notations used to formulate the HRES design problem. Note that square brackets denote index sets, e.g., $[T] := \{1, 2, \dots, T\}$. We first present the deterministic model, so the subscript s is suppressed in all parameters and variables, but will be used later in the stochastic case. The first set of constraints (1) links the design variables to the state and control variables by capping the charging and discharging powers and the energy stored in the BB based on the number of battery modules installed. It also stipulates that the power delivered by the DG cannot exceed its rated capacity.

$$\begin{aligned}
 Dch_t &\leq MR_{bat} N_{bat} & \forall t \in [T] \\
 Ch_t &\leq MR_{bat} N_{bat} & \forall t \in [T] \\
 SoC_t &\leq SC_{bat} N_{bat} & \forall t \in [T] \\
 P_t^{DG} &\leq Cap_{DG} & \forall t \in [T]
 \end{aligned} \quad (1)$$

Constraint (2) enforces the energy conservation law in the BB, which states that the change in the energy quantity stored in it over a time period equals the difference between charged and discharged quantities (adjusted for charging cycle efficiency) during this period.

$$SoC_t - SoC_{t-1} = \Delta \left(Ch_t - \frac{Dch_t}{\eta_{bat}} \right) \quad \forall t \in [T] \quad (2)$$

The next constraint (3) links the supply and demand in every time period, ensuring that the total power extracted from all sources (PV, WT, DG and the power discharged from the BB) is greater than the power used (net demand, i.e., demand less curtailed loads, and the power used to charge the BB), where the difference (if any) represents the power dumped.

$$N_{PV} P_t^{PV} + N_{WT} P_t^{WT} + Dch_t + P_t^{DG} \geq d_t - U D_t + Ch_t \quad \forall t \in [T] \quad (3)$$

Finally, the following non-negativity and integrality constraints are added.

$$Dch_t, Ch_t, P_t^{DG}, SoC_t, U D_t \geq 0 \quad \forall t \in [T] \quad (4)$$

$$Cap_{DG}, N_{PV}, N_{WT}, N_{bat} \geq 0, \text{ integer} \quad (5)$$

Two objectives are sought: minimizing cost and maximizing reliability. In particular, the first objective function (6) encompasses the annualized costs of the system components and the fuel cost of the DG, and is referred to hereinafter as the *Total Annual Cost* (TAC). The first term represent the installation costs of the PV, WT and DG, annualized using the parameter $A = \frac{ir}{1-(1+ir)^{-n}}$, where ir and n are, respectively, the interest rate and the HRES lifetime. The second term represents the BB cost, calculated based on the assumption that battery modules can function for a specific number of full charging cycles (L_{bat}) before replacement. Specifically, the initial cost of a battery module is divided by its usable storage capacity and the lifetime number of charging cycles to compute the cost per kWh charged to the BB, which is then multiplied by the total energy charged over a year to obtain the annual cost of the BB. The last term represents the fuel cost of the DG, which is proportional to the total energy delivered by it over a year.

$$\begin{aligned}
 TAC = & A (C_{PV} N_{PV} + C_{WT} N_{WT} + C_{DG,1} Cap_{DG}) \\
 & + \frac{\Delta C_{bat}}{SC_{bat} L_{bat}} \sum_{t \in [T]} Ch_t + C_{DG,2} \sum_{t \in [T]} P_t^{DG}
 \end{aligned} \quad (6)$$

The second objective function (7) to be minimized measures the “unreliability” of the system. The LLP, which is a widely-used reliability index in HRESs [15,28,29], is defined as the fraction of total demand the system cannot satisfy because of insufficient energy supply, i.e., total curtailed load divided by total demand.

$$LLP = \frac{\sum_{t \in [T]} U D_t}{\sum_{t \in [T]} d_t} \quad (7)$$

Table 1

Notations.

Indices/Sets	
$t \in [T]$	Time periods
$s \in [S]$	Scenarios
$r \in [R]$	Design variables
Parameters	
d_t	Power demand during time period t (kW)
$P_{t,s}^{PV}$	Power output from a PV panel during time period t in scenario s (kW)
$P_{t,s}^{WT}$	Power output from a wind turbine during time period t in scenario s (kW)
q_s	Nominal probability of scenario s
SC_{bat}	Usable energy storage capacity of a battery module (kWh)
MR_{bat}	Maximum charge/discharge rate (power) of a battery module (kW)
$C_{DG,1}$	Capacity cost of a diesel generator (\$/kW)
$C_{DG,2}$	Operating cost for a diesel generator (\$/kWh)
C_{PV}	Unit cost of a PV panel (\$/Unit)
C_{WT}	Unit cost of a wind turbine (\$/Unit)
C_{bat}	Unit cost of a battery module (\$/Unit)
L_{bat}	Lifetime number of full charging cycles of a battery module
η_{bat}	BB charging cycle efficiency
Δ	Duration of each time period
α	Penalty factor for unmet demand (\$/kWh)
ir	Interest (discount) rate
n	Useful life of the HRES (year)
Design Variables	
N_{PV}	Number of PV panels
N_{WT}	Number of wind turbines
N_{bat}	Number of battery modules
Cap_{DG}	Rated capacity of the diesel generator (kW)
Control Variables	
$Dch_{t,s}$	Power discharge from the BB during time period t under scenario s (kW)
$Ch_{t,s}$	Power charge to the BB during time period t under scenario s (kW)
$P_{t,s}^{DG}$	Power supplied by the DG during time period t under scenario s (kW)
State Variables	
$SoC_{t,s}$	BB State-of-charge at the end of time period t under scenario s (kWh)
$UD_{t,s}$	Unmet power demand during time period t under scenario s (kW)

For the purpose of applying the proposed solution methods, the bi-objective HRES design problem is transformed into a single cost-minimizing problem through multiplying LLP by $\alpha\Delta \sum_{t \in [T]} d_t$ and adding it to TAC. This is equivalent to penalizing unmet demand at a rate of α /kWh. It should be noted that, under very mild assumptions about the problem parameters (e.g., $\alpha > C_{DG,1}$), the mathematical formulation provided above enforces the operational rules described in the previous section. We also note that the system cost does not depend on the initial number of battery modules installed, but rather on the rate at which they are used (i.e., charged and discharged). Hence, to avoid having an unnecessarily large BB (beyond what is needed in constrain set (1)), we penalize N_{bat} using a small artificial cost that is added to TAC when solving the problem.

3. Modeling uncertainty

The mathematical model presented in the previous section treats PV and WT outputs as deterministic parameters. In reality, however, these are time-series data characterized by high variability, uncertainty and auto-correlation [30]. To capture these characteristics, and given the daily cyclic nature of RES supply data, one can simply consider each available supply sample (i.e., day) as a “scenario” and optimize the expected objective value, where the expectation is taken with respect to the distribution of supply scenarios. However, the problem becomes computationally expensive when the number of scenarios is large, which calls for scenario reduction techniques to be applied. Hence, we use k -means clustering to extract a manageable number of supply scenarios that can reasonably model parameter variability while maintaining tractability. Furthermore, Since the probability distribution of the extracted scenarios is based a relatively small number of sample points, it can significantly deviate from the true distribution. Thus, we employ a robust approach to hedge the HRES design against distributional ambiguity. A brief description of the DAS utilized for this purpose is provided later.

3.1. k -Means clustering to extract supply scenarios

A k -means clustering algorithm iteratively tries to partition the dataset, which consists of $N = 365$ sample points, into k non-overlapping subgroups (in this study, S scenarios). It is based on the minimization of an objective function defined as [31]:

$$\min \sum_{s \in [S]} \sum_{n \in [N]} \mu_{ns} \|x_n - v_s\|_2, \quad (8)$$

where x_n is the actual power output in day n and v_s is the center of scenario s . Also μ_{ns} is the assignment variable which equals 1 if the daily profile n is assigned to cluster s and 0 otherwise. Both x_n and v_s are vectors consisting of 24 elements (24 h per day). This optimization problem can be solved using a heuristic iterative algorithm explained as follows [31]:

1. Specify the number of clusters/scenarios (S) and initialize their centroids (v_s , $s \in [S]$) by drawing randomly from the element-wise intervals $[x_n^{\min}, x_n^{\max}]$.
2. In iteration j :
 - Calculate the distance $d_{ns}^j = \|x_n - v_s^j\|_2$ between each sample point (x_n) and cluster centroid (v_s^j).
 - Tentatively assign each sample point to the cluster with the nearest centroid to it, i.e., set $\mu_{n\bar{s}}^j = 1$, where $\bar{s} = \operatorname{argmin}_{s \in [S]} d_{ns}^j$ and all other μ_{ns}^j , $s \in [S]/\{\bar{s}\}$ to 0.
 - Update the coordinates of each cluster's centroid to coincide with the arithmetic mean of the sample points assigned to it, as follows:

$$v_s^{j+1} = \frac{\sum_{n \in [N]} \mu_{ns}^j x_n}{\sum_{n \in [N]} \mu_{ns}^j} \quad \forall s \in [S]. \quad (9)$$

- Keep repeating step 2 until the change in the sum of squared distances is less than a certain tolerance ϵ :

$$\left| \sum_{s \in [S]} \sum_{n \in [N]} (d_{ns}^{j+1} \mu_{ns}^{j+1})^2 - \sum_{s \in [S]} \sum_{n \in [N]} (d_{ns}^j \mu_{ns}^j)^2 \right| \leq \epsilon. \quad (10)$$

- Calculate the probability of each scenario as:

$$q_s = \frac{\sum_{n \in [N]} \mu_{ns}}{N} \quad \forall s \in [S]. \quad (11)$$

Clearly, this is a heuristic that does not guarantee global optimality for the non-convex k -means clustering problem. To reduce the likelihood of converging to a local optimum, the algorithm is repeated several times starting from different initial cluster centroids, and the best clustering output (*i.e.*, one with the least within-cluster variance) is declared optimum. Moreover, the clustering output improves as the number of clusters increase. Thus, we begin with $S = 2$ and increase the number of clusters incrementally until the marginal reduction in within-cluster variance drops below a specific threshold.

3.2. ϕ -divergence ambiguity set

To represent the ambiguity about the probability distribution of the scenarios extracted in the previous section, we construct a DAS based on ϕ -divergence, similar to that introduced in [32]. A ϕ -divergence measures the difference between a pair of non-negative vectors, $p = (p_1, \dots, p_S)^T$ and $q = (q_1, \dots, q_S)^T$ where p is the unknown true probability distribution and q is the empirical estimate of p . Formally, a ϕ -divergence is defined as

$$I_\phi(p, q) = \sum_{s \in [S]} q_s \phi\left(\frac{p_s}{q_s}\right), \quad (12)$$

where $\phi(t)$ is referred to as the ϕ -divergence function and is convex for $t \geq 0$ with $\phi(1) = 0$. The ambiguity set for p defines the family of distributions

$$P_\phi := \left\{ p \mid p \geq 0, \sum_{s \in [S]} p_s = 1, I_\phi(p, q) \leq \rho \right\}, \quad (13)$$

i.e., all distributions supported on the extracted scenarios and have divergence values of at most ρ from the empirical distribution.

Most ϕ -divergences do not satisfy the triangle inequality, and many are not symmetric in the sense that $I_\phi(p, q) \neq I_\phi(q, p)$ [33]. However, one exception is the *Variation Distance* (VD), defined as $I_\phi(p, q) = \sum_{s \in [S]} |p_s - q_s|$, or simply the l_1 -norm of the difference between the two vectors. Besides being a linear function, thus preserving tractability, using this divergence metric enables “suppression” and “popping” of scenarios, *i.e.*, allowing, respectively, $p_s = 0$ for a scenario for which $q_s > 0$, and $p_s > 0$ for a scenario for which $q_s = 0$ [33]. Also, as shown in [34], using VD as the ϕ -divergence makes the problem solvable using efficient decomposition algorithms. Thus, this is the ϕ -divergence of choice herein to construct the DAS.

The parameter ρ plays a key role in determining the quality of the obtained solution. Setting $\rho = 0$ leads to a unitary DAS that includes only the empirical distribution q , thus reduces the problem to the SAA case, which is prone to over-fitting. At the other extreme, setting $\rho = 2$ (for a VD-based DAS) ensures that all the probability mass can be moved from some scenarios to others, *e.g.*, the entire probability mass is placed on scenario s_1 in p and on scenario s_2 in q . Using an oversized DAS (by setting ρ too large) admits many irrelevant distributions and leads to solutions that are overly-conservative and under-performing. The “optimal” value of ρ , which leads to the best expected performance, is usually small, and depends on the variance of the uncertain parameters, size of the data sample used to construct the empirical distribution, the number of scenarios, and the desired

confidence level. This optimal value can be determined through *cross-validation*, by which the solutions obtained using different ρ values are simulated on new (*i.e.*, out-of-sample) data and the ρ value corresponding to the best-performing solution is considered optimum. The cross-validation method used to properly size the DAS is explained in more detail and implemented in Sections 5.2 and 5.3.

4. Solution methods

Unlike the deterministic model where explicit functions exist to evaluate TAC and LLP under simplifying assumptions, such closed-form expressions do not exist for the real-case (*i.e.*, with real RES supply data). Instead, these objectives have to be estimated using simulation and optimized in a SO framework. In what follows, we present two metamodel-based SO approaches that successively approximate the complex I/O surface of the simulation model using easy-to-optimize surrogate functions and show how to deal with distributional ambiguity in both.

4.1. Classical response surface methodology

RSM is a widely-used, step-wise heuristic method that estimates the best inputs combination which can minimize a given objective function [35]. In this study, a robust version of the classical RSM is proposed along with its nominal version. The RSM description provided below is based on [36].

Initialization: Select a starting point (HRES configuration) based on intuition, prior knowledge, or the current system configuration.

Stage 1: For the current solution X , do:

- Estimate the combined objective value $Y_s(X) = TAC_s + \alpha \Delta \sum_{t \in [T]} LLP_s$ for each scenario $s \in [S]$ through simulation.
- Find the worst-case distribution p by solving

$$\max_{p \in P_\phi} \sum_{s \in [S]} Y_s(X) p_s. \quad (14)$$
- Explore the I/O behavior of the simulation in the neighborhood of X using a Resolution-III (R-III) design.
- Fit the first-degree polynomial $\hat{Y}_s = \beta_{0,s} + \sum_{r \in [R]} \beta_{r,s} X_r$ for each scenario $s \in [S]$ based on X and the points evaluated around it using the least squares method. Note that $X = \{X_r\}_{r \in [R]} = \{N_{pv}, N_{wt}, N_{bat}, Cap_{DG}\}$. Compute the weighted average regression plane $\hat{Y} = \sum_{s \in [S]} p_s \hat{Y}_s$.
- Find the gradient $\nabla = [\beta_1, \beta_2, \beta_3, \beta_4]$. If $\|\nabla\|_\infty < \epsilon$, skip to Stage 2.
- Perform a path search along the steepest descent direction $-\nabla$ to identify t that minimizes $\hat{Y}$. Set $X \leftarrow X - \nabla t$ and go to step 1.

Stage 2: In the neighborhood of X :

- Explore the I/O behavior of the simulation in the neighborhood of X using a Central Composite design (CCD).
- Fit the second-degree polynomial $\hat{Y}_s = \beta_{0,s} + \sum_{r \in [R]} \beta_{r,s} X_r + \sum_{r,r' \in [R]} \beta_{rr',s} X_r X_{r'}$ for each scenario $s \in [S]$ based on X and the points evaluated around it using the least squares method. With four design variables, 15 coefficients β are estimated for the quadratic polynomial. Compute the weighted average regression paraboloid $\hat{Y} = \sum_{s \in [S]} p_s \hat{Y}_s$.
- Apply the first-order optimality conditions on the quadratic approximation to find X^* that minimizes $\hat{Y}$. Round it to the nearest integer number, and declare it as the optimal solution. Using simulation, estimate the optimal value $\hat{Y}^*$.

When solving the nominal problem (*i.e.*, without distributional ambiguity), the empirical distribution q is used directly instead of the worst-case distribution p when calculating the weighted average linear and quadratic regression surfaces. A pseudocode for the classical

RSM described above is presented in Algorithm 1. Furthermore, a graphical (flowchart) representation of the algorithm is provided in the [Appendix](#).

read: Solar, wind, and demand power data for all scenarios;

input : q, ϵ

output: Optimal value ($\hat{Y}^*$) and optimal solution (X^*)

Initialize starting points X ;

Initialize gradient;

while $\text{gradient} \leq \epsilon$ **do**

foreach scenario $s \in [S]$ **do** Calculate

 TotalAnnualCost (starting points);

 Find the worst-case distribution p by solving (14) (for the nominal case, set $p = q$);

 Generate a set of sample points with R-III design;

for each scenario $s \in [S]$ **do**

 Calculate TotalAnnualCost function evaluation;

 Fit a first-degree polynomial;

end

 Calculate the weighted average first-degree equation using p ;

 Apply steepest descent to get a new combination $\bar{X}$;

 Update gradient;

$X \leftarrow \bar{X}$;

end

Generate a set of sample points with CCD;

for each scenario $s \in [S]$ **do**

 Calculate TotalAnnualCost function evaluation;

 Fit a second-degree polynomial;

end

Calculate the weighted average second-degree equation;

Solve the second-degree polynomial to get the optimal HRES combination X^* and $\hat{Y}^*$

Algorithm 1: Classic Response Surface Methodology

4.2. Global response surface technique

In addition to the classical RSM described in the previous section, we propose a GRST to solve the problem under consideration. Unlike the RSM that locally approximates the response function in the neighborhood of the current solution and uses gradient information to iteratively move to better solutions, the GRST is an exploratory global approximation model that fits a second-degree polynomial on the entire search space from the beginning then refines the search space. The GRST is described as follows:

Initialization: Define a sufficiently large initial search space $\mathcal{X} := \{X \in \mathbb{N}_+^R : X_r \in [X_r^l, X_r^u], \forall r \in [R]\}$ that is known to include the optimal solution. Set $X^* = X^u$.

Iteration:

1. Generate a set of sample points using the Latin Hypercube Sampling (LHS) method from the current search space $\mathcal{X}$. Reuse the points in $\mathcal{X}$ carried forward from previous iterations. Evaluate the newly generated sample points through simulation for each supply scenario $s \in [S]$ to obtain $Y_s(X)$.
2. Fit a convex quadratic surrogate function $\hat{Y}_s(X)$ for each scenario based on the sample points by solving the least-squares regression quadratic optimization problem

$$\min_{\beta} \left\| Y_s(X) - \hat{Y}_s(X) \right\|_2^2 \quad (15)$$

$$\text{s.t.} \quad \mathbb{H}(\hat{Y}(X)) \succeq 0, \quad (16)$$

where the quadratic surrogate function is defined as

$$\begin{aligned} \hat{Y}(X) = & \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_1 X_2 + \beta_6 X_1 X_3 \\ & + \beta_7 X_1 X_4 + \beta_8 X_2 X_3 + \beta_9 X_2 X_4 + \beta_{10} X_3 X_4 + \beta_{11} X_1^2 \\ & + \beta_{12} X_2^2 + \beta_{13} X_3^2 + \beta_{14} X_4^2, \end{aligned}$$

and its Hessian matrix

$$\mathbb{H}(\hat{Y}_s(X)) = \begin{bmatrix} \beta_{11} & \beta_5 & \beta_6 & \beta_7 \\ \beta_5 & \beta_{12} & \beta_8 & \beta_9 \\ \beta_6 & \beta_8 & \beta_{13} & \beta_{10} \\ \beta_7 & \beta_9 & \beta_{10} & \beta_{14} \end{bmatrix} \quad (18)$$

must be positive semi-definite.

3. In the nominal case, solve the linearly-constrained convex quadratic optimization problem

$$\min_{X \in \mathcal{X}} \hat{Y}(X) = \sum_{s \in [S]} q_s \hat{Y}_s(X), \quad (19)$$

which minimizes the weighted average of the S fitted surrogate functions whose weights are the empirical probabilities q of the supply scenarios, to obtain X^* .

4. In the robust case, solve the min-max problem $\min_{X \in \mathcal{X}} \max_{p \in P_\phi} \sum_{s \in [S]} p_s \hat{Y}_s(X)$, which finds the solution X^* that minimizes the weighted average of the fitted surrogate functions whose weights are the probabilities from the worst-case distribution in the ϕ -divergence DAS defined based on VD. This problem can be expanded to

$$\begin{aligned} \min_{X \in \mathcal{X}} \max_{p \geq 0} \sum_{s \in [S]} p_s \hat{Y}_s(X) \\ \text{s.t.} \quad \sum_{s \in [S]} p_s = 1 \\ \sum_{s \in [S]} |p_s - q_s| \leq \rho. \end{aligned} \quad (20)$$

To tractably reformulate this problem, we first linearize the absolute value term by introducing the new variable $\gamma_s = |p_s - q_s|$, then apply linear programming (LP) duality to the inner maximization and combine it with the outer minimization over X to obtain the convex quadratic optimization problem

$$\begin{aligned} \min_{X \in \mathcal{X}, \delta, \eta, \pi^+, \pi^-} \quad & \eta + \rho \delta + \sum_{s \in [S]} q_s (\pi_s^+ - \pi_s^-) \\ \text{s.t.} \quad & \hat{Y}_s(X) - \eta - \pi_s^+ + \pi_s^- \leq 0 \quad \forall s \in [S] \\ & -\delta + \pi_s^+ + \pi_s^- \leq 0 \quad \forall s \in [S] \\ & \delta, \pi_s, \alpha_s \geq 0 \quad \forall s \in [S]. \end{aligned} \quad (21)$$

5. After obtaining and rounding X^* to the closest integer number, if $\|X^* - X^t\| < \epsilon$, terminate and declare X^* as the optimal solution. Using simulation, estimate the optimal value $\hat{Y}^*$.
6. Recenter the search space at X^* and shrink its size by half in each dimension. Carry forward any sample points situated in the new search space to the next iteration.

The pseudocode for the GRST described above is presented in Algorithm 2. Furthermore, a graphical (flowchart) representation of the algorithm is provided in the [Appendix](#).

5. Case study and results

The proposed methods were tested on a case study of a hypothetical remote community with a population of 50,000 situated in northwestern Ontario, Canada. Hourly demand data for one year was based on the power consumption of Northwest Ontario as reported in the Independent Electricity System Operator's (IESO) 2019 zonal demand report [37], prorated for the community population. Hence, the community has a peak and average load demands of 734 kW and 505 kW, respectively. Solar and wind power outputs were extracted from the IESO's *Generator Output and Capability Reports* [38], which provides hourly output levels grouped by fuel type, with a maximum output capability of 50 kW and 1 kW for wind turbines and PV panels, respectively. Grid-scale Sodium Sulfur (NaS) battery modules, a high-temperature rechargeable battery that uses sodium for the negative and

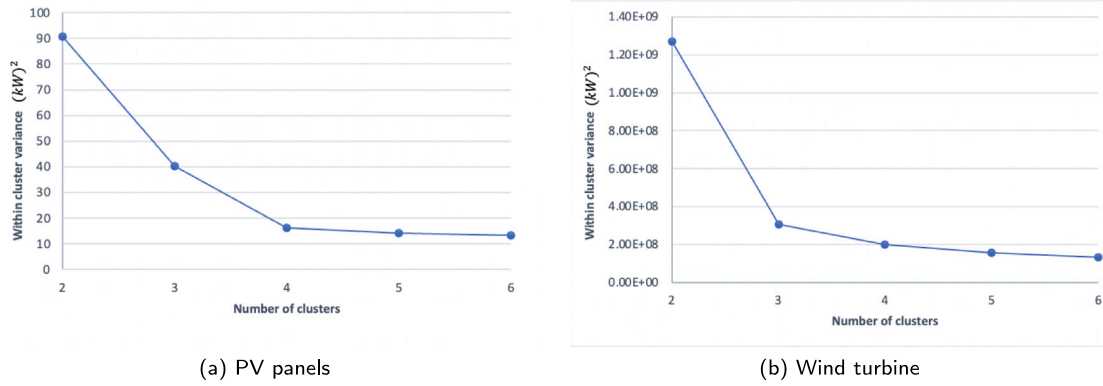


Fig. 2. Within-cluster variance vs. number of clusters.

Read: Solar, wind, and demand power data for all scenarios;

Input : q, ε

Output: $\hat{Y}^*, X^*$

Initialize the search space as

$$\mathcal{X} := \{X \in \mathbb{N}_+^R : X_r \in [X_r^l, X_r^u], \forall r \in [R]\};$$

Set $X_r^l = \frac{X_r^l + X_r^u}{2}$ and $X_r^d = \frac{X_r^u - X_r^l}{2}$;

Set $X^* = X^u$

while $\|X^* - X^l\| \geq \varepsilon$ **do**

$X^* \leftarrow X^l$

 Generate a set of sample points using LHS from $\mathcal{X}$;

for each scenario $s \in [S]$ **do**

 Calculate TotalAnnualCost function evaluation;

 Fit a second-degree surrogate function;

end

 Solve the nominal problem as in (19) and robust problem as in (21);

$$X_r^d \leftarrow \frac{X_r^d}{2};$$

 Update X_r^l and X_r^u as $X_r^l = \max(0, X_r^l - X_r^d)$ and $X_r^u = X_r^l + X_r^d$;

end

Report the optimal HRES combination X^* and its corresponding $\hat{Y}^*$

Algorithm 2: Global Response Surface Technique

sulfur for the positive electrode, are used for energy storage. Furthermore, the HRES uses a diesel generator to fill the gap between supply and demand beyond the battery bank capacity. Other parameters used to calculate the TAC are depicted in Table 2. Setup and operating costs of most system components (PV, WT and DG) were obtained from a local supplier [39], whereas the BB cost and characteristics were sourced from its manufacturer [40].

5.1. Generation of supply scenarios

Daily PV and WT supply scenarios are generated from historical data. Each scenario represents a full-day's supply profile of a PV panel or a wind turbine. Although it is possible to consider each daily supply profile over a year (or several years) as a separate scenario, this would result in an extremely large number of scenarios, exactly $(365)^2$, for a one-year data set, making the problem intractable. Hence, k -means clustering is utilized to extract a manageable number of scenarios that can capture the data variability reasonably well. To determine this number, we use the well-known elbow method, which increases the number of scenarios gradually until the incremental reduction in the within-cluster variance drops below a certain threshold (set here at 10%). Fig. 2 shows the within-cluster variance corresponding to different numbers of scenarios for PV panels and wind turbines. To avoid being

Table 2

Model parameters.

Parameter	Unit	Value
C_{PV}	\$/kW	2000
C_{WT}	\$/kW	2200
C_{bat}	\$/kW	1000
$C_{DG,1}$	\$/kW	350
$C_{DG,2}$	\$/kWh	0.182
SC_{bat}	kWh	2
MR_{bat}	kW	0.33
L_{bat}	Full cycle	4500
n	years	20

trapped in local optima, the clustering procedure was repeated 1000 times with different initial clusters, and the best among them is used for each case. We found that the improvement in clustering quality diminishes below the specified threshold as the numbers of PV and WT clusters are increased beyond 4 and 3, respectively; thus, these are the scenario numbers used. Fig. 3 illustrates the extracted PV and WT supply scenarios (i.e., cluster centroids). The marginal probability of each supply scenario is calculated by dividing the cluster size by 365. Moreover, assuming independence between the PV and WT scenarios, 12 combined supply scenarios are generated and their joint probabilities are computed as products of the marginal probabilities. Next, to generate annual supply profiles (to be used in the simulation), 365 daily supply profiles (scenarios) are drawn randomly with replacement according to the underlying probability distribution, whether nominal or robust. Table 3 depicts the marginal probabilities of supply scenarios.

5.2. Classical RSM results

When applying the classical RSM, problem (14) is solved using Gurobi, whereas the final quadratic function is optimized via its first-order optimality conditions. The design factors and levels considered for the R-III and CCD designs are presented in Tables 4 and 5, respectively. As an example for the R-III design, if the starting point is $(N_{PV} = 200, N_{WT} = 40, N_{bat} = 650, Cap_{DG} = 600)$, the minimum and maximum perturbed levels would be (190, 38, 520, 585) and (210, 42, 715, 630), respectively. Similar calculations are made for the CCD.

The results obtained from optimizing each scenario individually using the classical RSM are summarized in Tables 6–8 for three penalty costs. These penalty costs were selected deliberately to demonstrate the trade-off between minimizing TAC and maximizing reliability, otherwise higher values always lead to a 100% reliability. The optimal configuration, best TAC, computational time required for every scenario, number of iterations in the steepest descent procedure, reliability and the step size used for every scenario in the steepest descent are reported in Table 6 for a penalty cost of \$40/kWh.

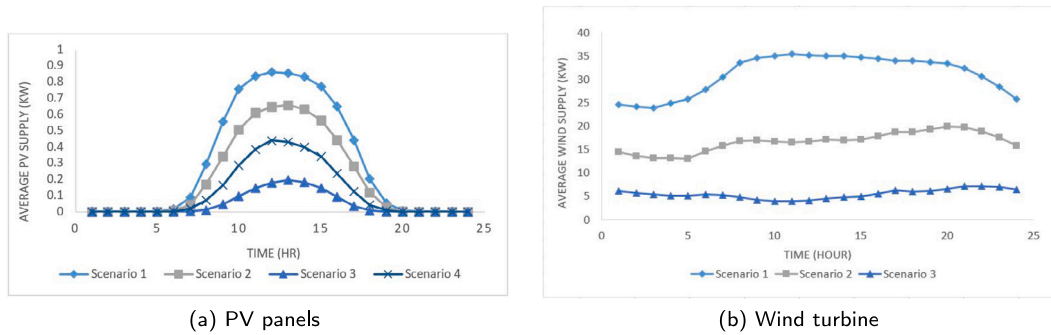


Fig. 3. Supply scenarios extracted from historical data (each scenario is a vector of size 24).

Table 3
Sizes and probabilities of the extracted clusters.

	Cluster size (PV)	Probability (PV)	Cluster size (WT)	Probability (WT)
Cluster 1	107	0.293	64	0.175
Cluster 2	85	0.233	129	0.353
Cluster 3	82	0.225	172	0.471
Cluster 4	91	0.249	–	–

Table 4
Factors and levels for the R-III design.

Design variables	Level	
	Minimum	Maximum
N_{PV}	–5%	+5%
N_{WT}	–5%	+5%
N_{bat}	–10%	+10%
Cap_{DG}	–5%	+5%

Table 5
Factors and levels for the CCD.

Design variables	Level				
	–2	–1	0	+1	+2
N_{PV}	–10%	–5%	0	+5%	+10%
N_{WT}	–10%	–5%	0	+5%	+10%
N_{bat}	–20%	–10%	0	+10%	+20%
Cap_{DG}	–10%	–5%	0	+5%	+10%

In the classical RSM approach, a starting point must be selected based on prior knowledge in order to initiate the process. Its selection depends on the intensity of input power data; therefore, it varies among different scenarios. Scenarios with high RES supply need fewer numbers of PV panels and wind turbines compared to those with low supply, and accordingly, they use smaller starting points. The number of iterations, computational time and the step size for each scenario are highly dependent on the starting point. The further the starting point is selected from the optimal configuration, the more time, number of iterations and larger step size (similar to step size of scenario #3 in Table 6) are needed to approach the final point. The step size in the last column is calculated using the method proposed in [41,42]. The optimal solution reported in the TAC column is without the “unreliability” penalty, i.e., for scenarios with a reliability less than 100%, the penalty cost is deducted from TAC and the actual cost is presented. According to Table 6, scenario #9, which uses the third cluster of both PV and WT power outputs, has the highest cost amongst all scenarios, whereas scenario #1, which uses the supply data from the first cluster of both PV panels and wind turbine power, has the lowest cost.

Choosing a high penalty cost (\$40/kWh) for unmet demand has led to 100% system reliability for almost all the scenarios since a large penalty must be incurred for supply deficiency. Thus, lower values (\$20/kWh and \$10/kWh) of the penalty cost are also tested and the results are presented in Tables 7 and 8. The same starting points that were used with the \$40/kWh penalty cost are used for these two cases. We notice that due to the lower value of penalty cost, fewer energy generation and storage units are installed and, accordingly, TAC has decreased while the system reliability has also deteriorated.

When assuming no ambiguity about the probabilities of scenarios, the best system configuration found using the classical RSM for the nominal problem consists of 404 PV panels, 27 wind turbines, 642 battery modules and a diesel generator with 655 kW capacity, resulting in a TAC of \$472,365 with \$40/kWh penalty cost and 99.99% reliability. As the penalty cost is decreased, the system tendency for allowing supply shortage increases, which leads to a TAC of \$452,041 for \$20/kWh penalty cost and 95.06% reliability, and a TAC of \$413,673 for \$10/kWh penalty cost and 84.53% reliability. A summary of the nominal problem results for different penalty costs is demonstrated in Table 9.

However, when there is uncertainty, it is important to size the DAS as small as possible but not too small. Setting $\rho = 0$ obtains the risk-neutral expected value minimization using the nominal distribution (i.e., the ϕ -divergence between p and q is zero). At the other extreme, as $\rho \rightarrow \infty$, the ambiguity set P_ϕ admits all possible distributions. The worst case among all possible distributions is to move all the probability mass to the scenario with the highest objective function value and set the probabilities of all the other scenarios to zero. This gives the most risk-averse approach and clearly shows the role of ρ as a risk-level parameter [33]. In this study, the highest value considered for ρ is 2, in which case all the probability mass is guaranteed to be moved to the scenario with the highest TAC. The idea is to select an ambiguity set to reflect the perceived risk from the data. Towards this purpose, cross-validation is performed using different values of ρ to determine the optimal TAC found based on different sizes of the DAS. Fig. 4 illustrates the cross-validation results for the three penalty costs.

As explained earlier, when $\rho = 0$ for a \$40/kWh penalty, TAC is \$472,521, which is quite close to the nominal problem (\$472,364), whereas as ρ approaches 2 (\$708,054 for $\rho = 1.8$), the ambiguity set admits all possible distributions and TAC becomes close to that of the worst-case scenario, scenario #9 in Table 6, with TAC of \$707,362. The same explanation goes to the cases with \$20/kWh and \$10/kWh

Table 6
Classical RSM — Optimal results for each scenario with \$40/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Time (s)	No. iteration	Reliability	Starting point	Step size
1 (1.1)	232	18	298	222	\$159,820	10.18	9	100%	[250,25,380,300]	1.4328e−03
2 (1.2)	272	33	402	183	\$270,923	10.02	9	100%	[300,40,500,250]	1.4337e−03
3 (1.3)	462	52	904	698	\$564,582	13.22	6	100%	[400,75,800,600]	0.07282
4 (2.1)	162	21	385	367	\$179,243	9.69	8	100%	[200,25,420,400]	1.4332e−03
5 (2.2)	236	35	337	373	\$289,021	10.73	13	100%	[300,45,400,400]	1.4317e−03
6 (2.3)	537	60	708	737	\$600,874	9.65	7	100%	[550,80,700,600]	5.0448e−03
7 (3.1)	242	20	244	353	\$180,134	10.50	12	100%	[300,30,300,400]	1.4291e−03
8 (3.2)	250	38	354	407	\$310,120	10.21	10	100%	[300,45,400,450]	1.4332e−03
9 (3.3)	687	81	1084	694	\$707,362	10.08	8	100%	[500,85,700,700]	4.8258e−03
10 (4.1)	263	19	265	326	\$181,212	10.07	8	100%	[300,25,300,350]	1.4332e−03
11 (4.2)	372	28	594	655	\$284,182	10.00	7	100%	[400,35,600,600]	1.8709e−03
12 (4.3)	611	70	768	396	\$638,476	10.83	11	99.22%	[600,80,650,850]	2.3730e−03

Table 7
Classical RSM — Optimal results for each scenario with \$20/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Time (s)	No. iteration	Reliability
1 (1.1)	130	13	148	68	\$138,577	13.11	14	95%
2 (1.2)	205	24	344	124	\$232,021	15.09	18	95%
3 (1.3)	376	46	776	432	\$561,172	10.20	12	93%
4 (2.1)	155	14	298	409	\$135,294	14.07	15	94%
5 (2.2)	204	25	405	238	\$263,735	17.21	23	93%
6 (2.3)	300	57	410	383	\$585,712	10.39	10	94%
7 (3.1)	209	13	202	328	\$177,075	15.21	19	94%
8 (3.2)	208	23	287	337	\$278,461	18.31	25	93%
9 (3.3)	230	49	442	527	\$691,383	19.43	28	95%
10 (4.1)	140	15	246	98	\$156,023	10.61	12	94%
11 (4.2)	300	23	347	236	\$272,124	11.09	12	95%
12 (4.3)	209	42	402	378	\$664,463	18.23	25	94%

Table 8
Classical RSM — Optimal results for each scenario with \$10/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Time (s)	No. iteration	Reliability
1 (1.1)	106	12	85	25	\$118,899	15.21	16	83%
2 (1.2)	170	21	220	65	\$213,762	16.26	20	84%
3 (1.3)	220	40	334	187	\$523,132	16.02	19	83%
4 (2.1)	123	13	386	20	\$129,249	12.74	14	84%
5 (2.2)	102	22	297	134	\$255,326	18.23	25	85%
6 (2.3)	232	55	318	135	\$538,153	11.83	12	84%
7 (3.1)	163	14	107	160	\$169,224	15.67	19	84%
8 (3.2)	140	21	228	148	\$276,822	19.68	27	86%
9 (3.3)	208	40	303	283	\$645,428	22.65	32	85%
10 (4.1)	38	13	136	45	\$148,130	12.25	14	83%
11 (4.2)	232	21	332	85	\$263,328	12.61	15	84%
12 (4.3)	141	30	413	256	\$625,923	21.05	30	85%

Table 9
Classical RMS — Nominal problem optimal cost and configuration.

Penalty cost	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Iteration	Time (s)	Reliability	Starting point	Step size
\$40/kWh	404	27	642	655	\$472,364	14	389.32	99.99%	[500,40,750,750]	2.85e-03
\$20/kWh	322	22	405	350	\$452,041	20	522.03	95.06%	[500,40,750,750]	2.85e-03
\$10/kWh	143	18	330	245	\$413,673	25	633.29	84.53%	[500,40,750,750]	2.85e-03

Table 10
Classical RSM — Robust problem optimal cost and configuration.

Penalty cost	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	No. iteration	Time (s)	Reliability
\$40/kWh	414	28	640	712	\$470,473	14	396.96	99.99%
\$20/kWh	257	22	450	347	\$451,138	21	546.38	94.66%
\$10/kWh	244	17	325	238	\$412,212	26	650.21	84.85%

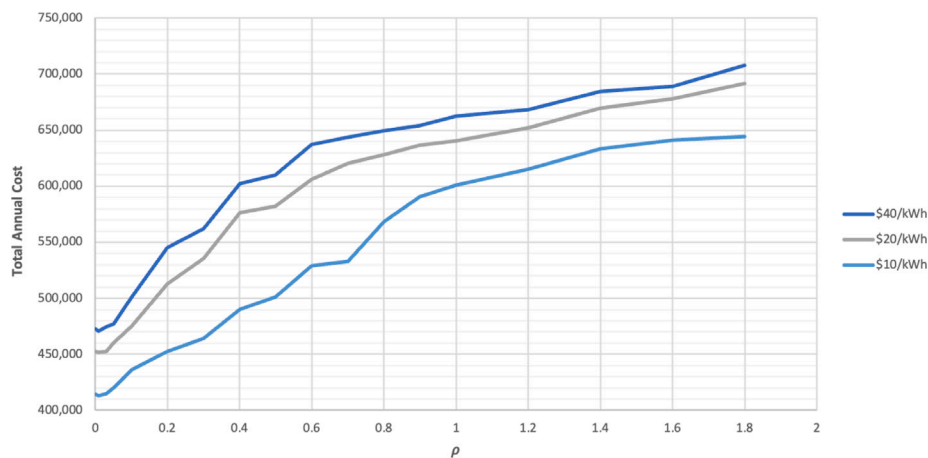


Fig. 4. Classical RSM: Cross-validation.

penalty costs. In the case with a \$20 penalty, when $\rho = 0$, TAC is \$452,276, which is close to the nominal problem's TAC of \$452,041. As ρ approaches 2 (\$690,834 for $\rho = 1.8$), TAC becomes close to that of the worst scenario, scenario #9 in Table 7, with TAC of \$691,383. The small difference between these values might have various explanations. It can happen because of the random DoE in every simulation iteration. Also, there is a possibility that the search converges to a local optimum in some cases.

If there is no ambiguity about the probability of scenarios, the nominal problem solution is the optimal. However, when there is ambiguity, uncertain parameters can randomly perturb the probabilities around their nominal values. Usually, a small value of ρ leads to a better out-of-sample performance than the nominal solution. In this study, considering a penalty cost of \$40/kWh, the best out-of-sample TAC found for the robust problem is when $\rho = 0.01$, leading to a TAC of \$470,473 with 414 PV panels, 28 wind turbines, 640 battery modules and a diesel generator with 712 kW capacity and 99% system reliability. The optimal solution achieved for $\rho = 0.01$ and \$20/kWh penalty cost is 257 PV panels, 22 wind turbines, 450 battery modules and a diesel generator with 347 kW capacity. TAC of this configuration is \$451,138 with 94.66% reliability. Optimal robust solutions achieved for different penalty costs are depicted in Table 10.

To confirm that $\rho = 0.01$ is the actual optimum value of ρ (i.e., it did not happen due to random sampling) and it outperforms the nominal solution ($\rho = 0$), we perform a paired t -test for $\rho = 0$ vs. $\rho = 0.01$ to determine which value of ρ (size of ambiguity set) gives a better expected out-of-sample performance and leads to a lower TAC. Therefore, the confidence interval for the difference between the

two expected responses is constructed to determine whether there is significant difference between TAC calculated with $\rho = 0$ and $\rho = 0.01$. $\mu_{0,j}$ and $\mu_{0.01,j}$ are paired together to define $Z_j = \mu_{0,j} - \mu_{0.01,j}$ for $j = 1, 2, \dots, n$. Z_j are i.i.d. variables and they are normally distributed. Thus the confidence interval for $E(Z_j)$ is calculated as follows:

$$\bar{Z}(n) = \frac{\sum_{j=1}^n Z_j}{n} \quad (22)$$

and

$$\widehat{Var}[\bar{Z}(n)] = \frac{\sum_{j=1}^n [Z_j - \bar{Z}(n)]^2}{n(n-1)} \quad (23)$$

We then form the (approximate) $100 \times (1 - \alpha)\%$ confidence interval as:

$$\bar{Z}(n) \pm t_{n-1, 1-\alpha/2} \sqrt{\widehat{Var}[\bar{Z}(n)]} \quad (24)$$

To assess the out-of-sample performance, 10 new random supply power realizations are generated similar to the original supply data in Section 5.1, i.e., 365 random data points are selected from both solar and wind clusters 10 times and new scenarios are used as simulation inputs to estimate the expected out-of-sample performance for the solutions obtained with different ρ values.

TAC values are depicted in Table 11 for \$40/kWh penalty cost. Assuming normality of the sample mean difference and $\alpha = 0.1$, a paired t -test is carried out. The obtained $\bar{Z}(10) = 1744.6$ and $\widehat{Var}[\bar{Z}(n)] = 51,593.6$, leading to the (approximate) 90% confidence interval [1328.2, 2160.9] for Z . Thus, with approximately 90 percent confidence, it can be concluded that μ_0 differs from $\mu_{0.01}$, and it furthermore appears that $\mu_{0.01}$ is superior, since it leads to a lower

Table 11
Classical RSM — TAC for 10 independent realizations with two $\rho = 0$ and $\rho = 0.01$, and their differences.

j	$\mu_{0,j}$	$\mu_{0.01,j}$	Z_j
1	472,922	470,102	2,820
2	472,180	470,232	1,948
3	472,831	469,927	2,904
4	471,555	470,520	1,035
5	472,374	470,396	1,978
6	471,899	471,123	776
7	472,911	471,021	1,890
8	472,893	471,262	1,631
9	471,795	470,714	1,081
10	472,207	470,824	1,383

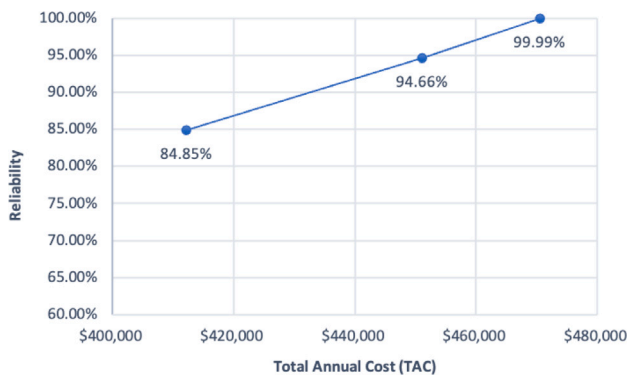


Fig. 5. Classical RSM — Pareto frontier for TAC vs. reliability.

TAC (between 1328.2 and 2160.9 lower). Based on the same normality assumption, similar calculations are done with new realizations for the penalty costs of \$20/kWh and \$10/kWh, resulting in confidence intervals for the mean differences of [964.6, 1645.5] and [1069.0, 1709.9], respectively, thus, likewise, indicating the superiority of $\rho = 0.01$.

In this study, when a high penalty cost is considered for unmet demand, the system attempts to install a large number of generation and storage units to avoid shortages, which enhances the reliability and increases the TAC. On the other extreme, by having a zero penalty cost, TAC can be reduced to zero by not installing any resources. However, the reliability index becomes zero, too. Therefore, the purpose is to look for “non-dominated” (or Pareto-optimal) solutions. The frontier in Fig. 5 is constructed to illustrate the trade-off between TAC and reliability by changing the penalty cost and observing how the cost and reliability change. Optimal TAC values obtained from the robust problem in Table 10 and their corresponding reliability values are drawn for three penalty costs in Fig. 8. Based on this Pareto Frontier, when a high value of penalty is considered for unmet demand, the system attempts to avoid any supply deficit and installs a large number of units, which leads to high system reliability and high TAC, accordingly. As the penalty cost decreases, the system allows more demand to remain unmet since supply deficiency is penalized at a lower rate.

5.3. GRST results

In performing the first step of the GRST, convexity of the estimated quadratic function is assured by using CVX [43] version 2.2, a Matlab-based modeling system for convex optimization. Each scenario is simulated and optimized using different penalty costs, and their optimal configuration, solution, computational time, number of iteration

and reliability are depicted in Tables 12–14 with \$40/kWh, \$20/kWh and \$10/kWh penalty costs, respectively. All the costs stated in the TAC column are actual costs, i.e., the penalty cost is deducted from TAC in the scenarios with reliability less than 100%. It can be inferred from Table 12 that scenario #1, which uses cluster 1 from both the PV and wind supply power, has the lowest TAC since fewer renewable resources are needed when a high amount of power is generated. On the contrary, scenario #9, which uses cluster 3 from both PV and wind as inputs, has the highest cost due to the large number of required resources. The same conclusion can be derived from Tables 13 and 14.

The GRST terminates when the difference between the lower and upper bounds of all the variables are negligible. Therefore, the number of iterations depends on the lower and upper limit of variables. To hold the same initial conditions for both methods, GRST upper bound and lower bound for each scenario are chosen based on the starting point of that scenario, i.e., the starting point of each scenario in the classical RSM is at the center of the starting range in the GRST. For example, if the starting point is [500, 40, 750, 750] in the classical RSM, the upper bound and lower bounds for GRST are set at [+20%, +25%, +20%, +20%] = [600, 50, 900, 900] and [−20%, −25%, −20%, −20%] = [400, 30, 600, 600], respectively.

Starting with a high penalty cost (\$40/kWh) for unmet demand in Table 12 has resulted in 100% reliability for most of the scenarios since high cost must be paid as penalty for supply deficiency. Next, smaller values (\$20/kWh and \$10/kWh) of the penalty cost are also tested and the results are presented in Tables 13 and 14. The same upper and lower bounds for each scenario are used for these two cases. Due to the lower penalty cost, fewer number of sources are installed, and accordingly, TAC has decreased and reliability has deteriorated. To solve the nominal problem, a convex quadratic function is fit on each scenario using Eq. (15) and empirical probabilities derived from clustering are used to obtain the optimal solution and configuration as in Table 15 for the three penalty costs.

Similar to the classical RSM, cross-validation is applied to determine the optimal ρ value in the GRST, and its results are demonstrated in Fig. 6 for the three penalty costs. For the first penalty cost of \$40/kWh, like the previous method, when $\rho = 0$, there is no ambiguity and the problem is similar to the nominal problem (\$472,565) with TAC of \$472,893. Whereas when ρ is approaching its maximum value of 2, the TAC becomes close to the worst scenario, scenario #9 (\$710,267), with TAC of \$709,234 ($\rho = 1.8$). Perturbing probabilities around their nominal values with small ρ , the best obtained TAC is with $\rho = 0.01$, which is \$471,327 with 421 PV panels, 25 wind turbines, 662 battery modules and a diesel generator with 695 kW capacity.

A similar explanation can be given regarding the solutions with \$20/kWh and \$10/kWh penalty costs. Due to their relatively lower costs, their plots situate below the solutions of \$40/kWh, and as the size of the ambiguity set increases, their TAC increases too. For a \$20/kWh penalty, when $\rho = 0$, TAC is \$453,293 (nominal problem = \$453,499) and as ρ gets closer to 2, the problem acts like the worst scenario (\$692,291) with a TAC of \$691,534 ($\rho = 1.8$). Moreover, with a penalty of \$10/kWh, TAC is \$414,823 and \$645,921 with $\rho = 0$ and $\rho = 1.8$, respectively. The nominal solution and the solution with $\rho = 0$, as well as the worst scenario solution and solution with $\rho = 2$, must coincide in theory. However, due to the random nature of GRST in every iteration of Monte Carlo simulation and the possibility of approaching a local optimum, there is a negligible difference (less than 0.001) between their values. The optimal solutions found for both robust problems are with $\rho = 0.01$, which has led to the lowest TAC. Best solutions and configurations found for the robust problem with the three penalty costs are summarized in Table 16.

A similar paired t-test is conducted to make sure there is a significant difference between the results of $\rho = 0$ vs. $\rho = 0.01$, and that $\rho = 0.01$ does give a better expected out-of-sample performance and leads to a lower TAC. Therefore, a confidence interval for the difference between the two expected response is constructed similar

Table 12
GRST — Optimal results for each scenario with \$40/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Computational time (s)	Iteration	Reliability
1 (1.1)	122	18	278	156	\$148,720	17.25	9	100%
2 (1.2)	290	33	460	186	\$272,907	13.57	8	100%
3 (1.3)	330	63	792	563	\$585,298	14.54	8	99%
4 (2.1)	154	18	390	303	\$160,030	13.69	8	100%
5 (2.2)	214	33	373	380	\$272,029	15.33	9	100%
6 (2.3)	518	68	680	585	\$632,415	13.65	8	99%
7 (3.1)	252	20	255	382	\$187,639	15.42	9	100%
8 (3.2)	218	35	340	427	\$293,317	15.11	9	100%
9 (3.3)	466	77	664	645	\$710,267	13.64	8	100%
10 (4.1)	263	18	278	340	\$174,194	15.17	9	100%
11 (4.2)	395	29	587	560	\$283,589	16.11	9	99%
12 (4.3)	227	60	590	718	\$697,880	15.26	9	100%

Table 13
GRST — Optimal results for each scenario with \$20/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Computational time (s)	Iteration	Reliability
1 (1.1)	90	14	154	80	\$139,478	13.73	8	94%
2 (1.2)	214	24	363	136	\$234,343	13.81	8	95%
3 (1.3)	294	50	450	263	\$562,062	12.01	7	93%
4 (2.1)	185	14	373	56	\$136,409	14.07	8	95%
5 (2.2)	102	25	332	195	\$263,084	15.82	9	95%
6 (2.3)	293	58	383	275	\$586,971	13.39	8	94%
7 (3.1)	152	14	253	143	\$176,440	15.21	9	95%
8 (3.2)	183	25	240	168	\$277,962	15.31	9	93%
9 (3.3)	256	48	430	355	\$692,291	13.44	8	94%
10 (4.1)	131	14	205	108	\$156,916	17.71	10	94%
11 (4.2)	267	24	337	163	\$271,225	15.09	9	94%
12 (4.3)	219	42	430	342	\$665,446	13.81	8	93%

Table 14
GRST — Optimal results for each scenario with \$10/kWh penalty cost.

Scenario # (PV#.WT#)	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Computational time (s)	Iteration	Reliability
1 (1.1)	66	12	86	30	\$121,546	14.02	8	82%
2 (1.2)	195	20	250	72	\$218,046	15.23	9	84%
3 (1.3)	214	39	384	205	\$525,281	13.86	8	83%
4 (2.1)	115	12	393	36	\$130,449	13.77	8	84%
5 (2.2)	93	21	302	114	\$254,232	15.22	9	85%
6 (2.3)	243	54	325	163	\$539,115	13.78	8	84%
7 (3.1)	193	12	127	60	\$168,884	15.07	9	84%
8 (3.2)	152	22	225	128	\$275,682	15.48	9	86%
9 (3.3)	215	40	310	270	\$646,401	13.76	8	84%
10 (4.1)	56	12	146	55	\$148,633	15.26	9	83%
11 (4.2)	264	20	305	106	\$264,681	15.01	9	84%
12 (4.3)	125	28	430	305	\$627,092	13.35	8	85%

to the previous method and $\mu_{0,j}$ and $\mu_{0.01,j}$ are paired together to define $Z_j = \mu_{0,j} - \mu_{0.01,j}$ for n new realizations. $\bar{Z}(n)$ and $\widehat{Var}[\bar{Z}(n)]$ are calculated according to Eqs. (22) and (23) and the (approximate) $100 \times (1 - \alpha)\%$ confidence interval is constructed using Eq. (24).

As done previously, 10 new random supply power realizations are generated similar to the original supply data and TACs are calculated in Table 17 with a penalty cost of \$40/kWh. Assuming normality of the sample mean difference (data are scattered near the straight line

Table 15
GRST — Nominal problem optimal cost and configuration.

Penalty cost	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Iteration	Computational time (s)	Reliability
\$40/kWh	432	26	662	683	\$472,565	11	199	99.99%
\$20/kWh	357	20	422	345	\$453,499	10	209	94.46%
\$10/kWh	235	17	352	256	\$414,603	11	222	85.55%

Table 16
GRST — Robust problem optimal cost and configuration.

Penalty cost	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Iteration	Computational time (s)	Reliability
\$40/kWh	421	25	662	695	\$471,327	10	478	100%
\$20/kWh	325	19	420	360	\$452,527	11	505	93.52%
\$10/kWh	218	16	345	275	\$413,527	10	476	84.72%

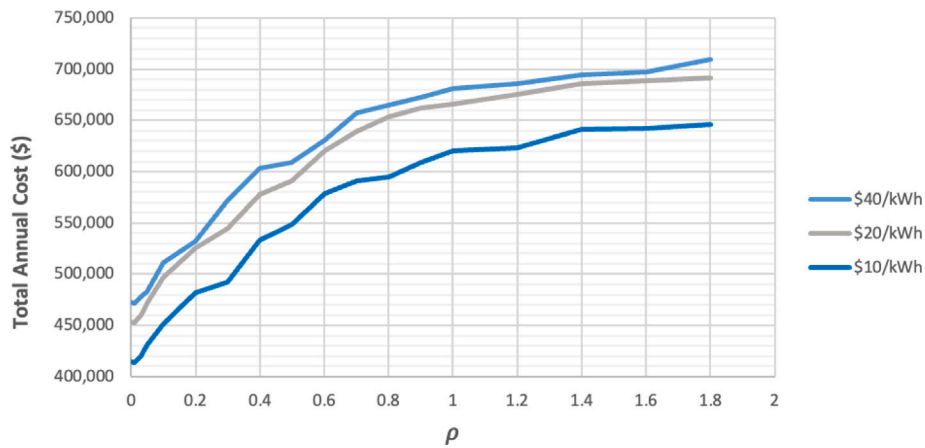


Fig. 6. GRST — Cross-validation.

Table 17
GRST — TAC for 10 independent realizations with two $\rho = 0$ and $\rho = 0.01$, and their differences.

j	$\mu_{0,j}$	$\mu_{0.01,j}$	Z_j
1	473,150	471,430	1720
2	473,926	472,665	1261
3	473,204	472,169	1035
4	472,810	471,576	1234
5	473,332	472,383	949
6	472,007	471,523	484
7	472,763	471,274	1489
8	473,075	471,551	1524
9	471,906	471,032	874
10	472,769	471,795	974

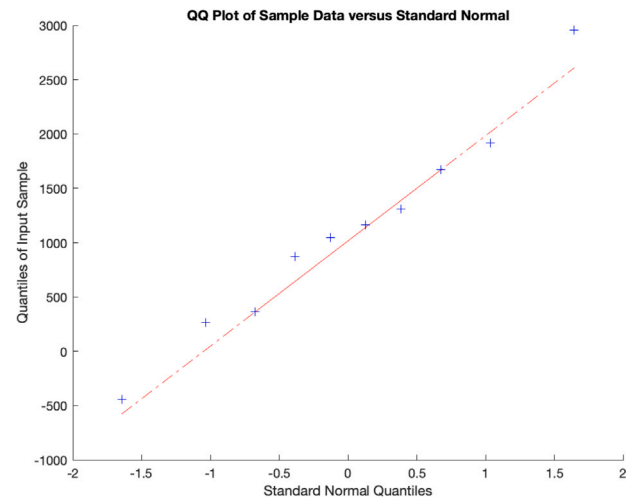


Fig. 7. GRST — QQ Plot for Z_j .

in Fig. 7) and $\alpha = 0.01$, a paired t-test is carried out. The obtained $\bar{Z}(10) = 1154.4$ and $\widehat{Var}[\bar{Z}(n)] = 13,369.5$ lead to the (approximate) 90% confidence interval $[942.4, 1366.3]$ for Z . Thus, with approximately 90 percent confidence, it can be concluded that μ_0 differs from $\mu_{0.01}$, and it furthermore appears that $\mu_{0.01}$ is superior, since it leads to a lower TAC (between 942.4 and 1366.3 lower). Based on the same normality assumption, similar calculations are done with new realizations for the penalty costs of \$20/kWh and \$10/kWh. The obtained confidence

intervals for Z are $[740.1, 1489.4]$ and $[836.2, 1545.3]$ for \$20/kWh and \$10/kWh, respectively, indicating the superiority of $\rho = 0.01$.

Next, a Pareto frontier is constructed to demonstrate the trade-off between TAC and reliability. Looking for the Pareto-optimal solutions, the optimal TAC deduced from the robust problem in Table 16 and their corresponding reliability values are plotted for three penalty

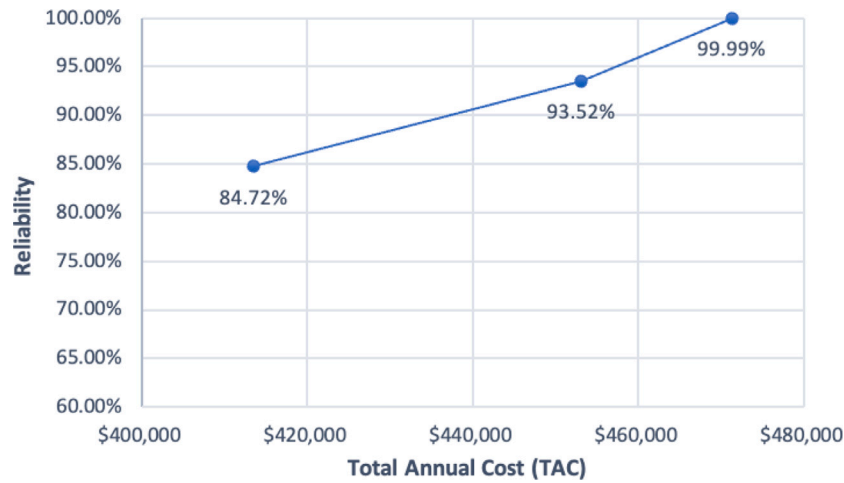


Fig. 8. GRST — Pareto frontier for TAC vs. reliability.

Table 18
Comparison between classical RSM and GRST results.

Problem	Penalty cost	N_{PV}	N_{WT}	N_{bat}	Cap_{DG} (kW)	TAC	Reliability	Computational time (s)
Nominal	\$40/kWh	6.9%	−3.7%	3.1%	4.3%	0.0%	0.0%	−48.9%
	\$20/kWh	10.9%	−9.1%	4.2%	−1.4%	0.3%	−0.6%	−60.0%
	\$10/kWh	−3.3%	−5.6%	6.7%	4.5%	0.2%	1.2%	−64.9%
Robust	\$40/kWh	1.7%	−10.7%	3.4%	−2.4%	0.2%	0.0%	20.4%
	\$20/kWh	26.5%	−13.6%	−6.7%	3.7%	0.3%	−1.2%	−7.6%
	\$10/kWh	−10.7%	−5.9%	6.2%	15.5%	0.3%	−0.2%	−26.8%

costs in Fig. 8. For a high penalty cost such as \$40/kWh, reliability of the system is near 100% since no unmet demand is allowed due to the high penalty added to TAC (\$471,327). As the penalty cost is decreased, the system allows some demand to remain unmet for lower TAC (\$453,127 and \$413,527), which also results in the lower reliability levels of 93.52% and 84.72% for \$20/kWh and \$10/kWh penalty costs, respectively. From the obtained solutions, the designers can choose the penalty costs they consider more appropriate to their utility.

5.4. Comparison between classical RSM and GRST

Given that two methods are proposed in this paper for designing HRESs, it is worth comparing them in light of the numerical results presented earlier. First, the optimal results obtained for both the nominal and the robust problems with different penalty costs are directly compared, then the pros and cons of each method are highlighted. Finally, system operation in several cases is simulated and analyzed.

Table 18 depicts the percentage differences between the optimal solutions (N_{PV} , N_{WT} , N_{bat} & Cap_{DG}), optimal values (TAC & Reliability) and computational times of the two methods for the nominal and the robust problem with \$40/kWh, \$20/kWh and \$10/kWh penalty costs. The reported percentage difference for a metric θ is computed as $\frac{\theta_{Classical\ RSM} - \theta_{GRST}}{\theta_{Classical\ RSM}}$, i.e., the difference is positive when $\theta_{Classical\ RSM} > \theta_{GRST}$, and vice versa.

We first note that the two methods led to substantially different solutions in all cases, with relative differences in component numbers/capacities ranging between 1.7% and 26.5%. However, the relative differences in TAC and reliability between the optimal designs obtained from the two methods are much smaller, not exceeding 0.3% and 1.2%, respectively. This observation demonstrates the ability of both methods to find high-quality, albeit different, solutions. The classical RSM slightly (yet consistently) outperforms the GRST in terms

of TAC, whereas the reliability comparison results are mixed with no clear advantage for either method. Careful examination of the optimal solutions reveals the substitutability of system components with others to achieve a similar performance. For example, in the nominal problem with \$40/kWh penalty cost, the GRST solution used 3.7% fewer WT than the classical RSM solution, but compensated for them by installing 6.9% more PV, 3.1% more BB capacity and 4.3% more DG capacity to achieve almost the same TAC and reliability. This substitutability explains why the problem might have multiple near-optimal, but very different, solutions. Furthermore, it is clear that the GRST enjoys much smaller computational times compared to the classical RSM when implemented to the nominal problem, with savings that range between 48.9% and 64.9%. In contrast, this computational time advantage of the GRST is inconclusive for the robust problem, perhaps because it requires the min-max problem (20) to be solved in every iteration.

Based on the comparative analysis performed above, the main advantage of the classical RSM over the GRST is its ability to find slightly better solutions in terms of TAC, and sometimes also reliability. However, its performance is found to be highly dependent on the starting point and the design variables' levels, i.e., R-III and CCD levels and intervals. A starting point that is too far from optimal or improperly set design levels might lead the search in a “wrong” direction based on local information, thus ending at a local optimum. In other words, to reach a near-optimal solution using the classical RSM, prior knowledge about the system is needed. On the other hand, although the solutions obtained using the GRST have marginally higher TAC values compared to the classical RSM solutions, the computational times for the latter are significantly smaller, especially in the nominal case. Furthermore, the number of iterations can be controlled by the planner via setting the initial search space and the stopping criterion (since the search space is reduced by a predetermined factor in each iteration). Another advantage of the GRST is that it uses global information from the

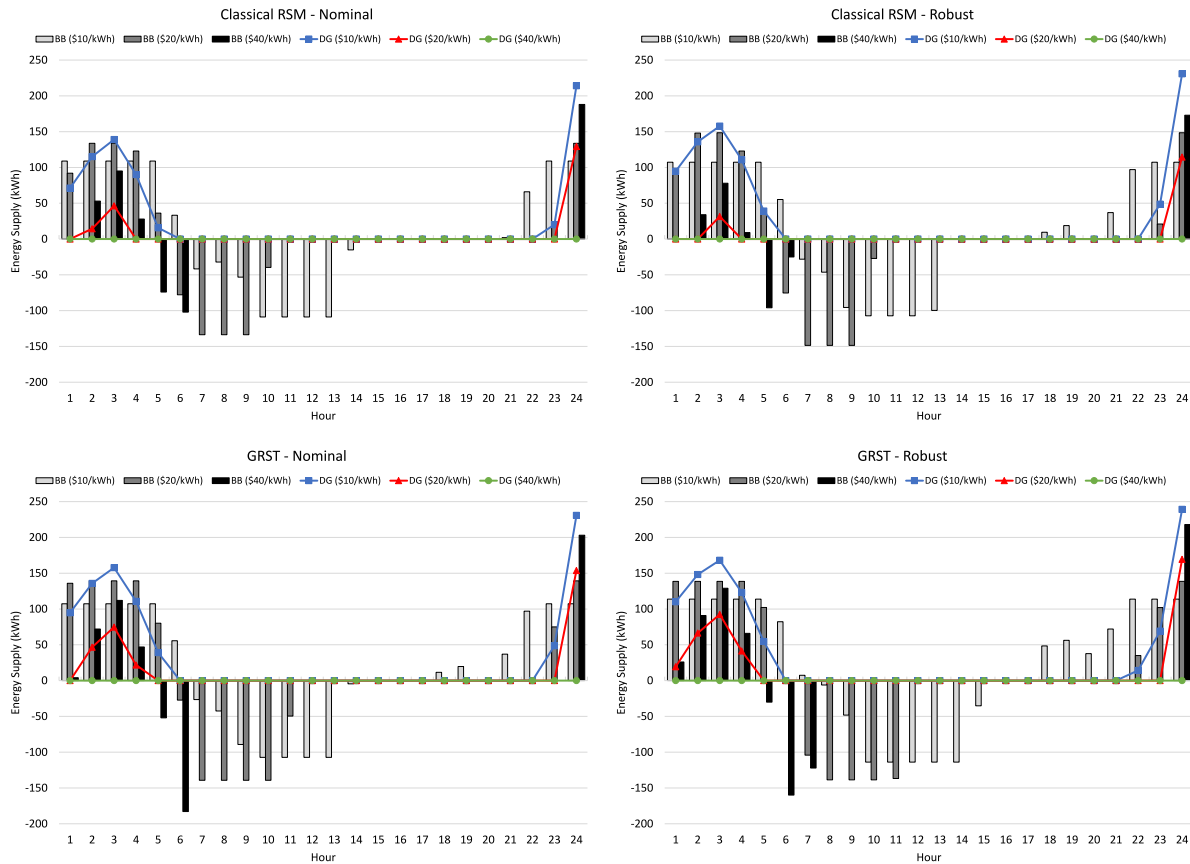


Fig. 9. HRES operation for one day.

entire search space in each iteration, thus is less sensitive to the initial point selection than the classical RSM. However, if the response surface cannot be reasonably approximated using a convex quadratic function, the accuracy of the GRST can severely suffer, thus reducing its likelihood of finding the global optimum. A possible remedy for this issue is by partitioning the search space into smaller areas and applying the GRST to each one independently. Moreover, increasing the number of sample points generated and simulated in each iteration can increase the approximation accuracy.

Finally, the optimal solutions obtained from the two methods with different penalty costs (\$10/kWh, \$20/kWh and \$40/kWh) for both the nominal ($\rho = 0$) and the robust ($\rho = 0.01$) cases are simulated using demand and RES supply data of the same day (1 December 2019), and the simulation results are presented in Fig. 9. The bar charts show the BB charging and discharging operations (negative and positive, respectively), whereas the DG power output is shown using the line charts. The BB and DG operation profiles for the day are quite similar across the 12 cases considered, in the sense that the BB is discharged and the DG is used during nighttime, and the BB is charged in the morning, which is expected given the PV supply profile. However, there are still some notable differences among the cases. In general, we note that the GRST solutions make slightly more use of the BB and DG to satisfy demand than the classical RSM solutions, which seem to rely more on their larger WT capacity. Likewise, the usage of the BB and DG in the robust solutions is slightly more than in the corresponding nominal solutions. The penalty cost seems also to play a significant role in determining how they are used. The smallest value of \$10/kWh is associated with the highest usage of the DG across the board, whereas the DG is not used at all when the penalty cost is \$40/kWh. This result is probably related to the high PV, WT and BB capacity installed in the

latter case, reducing the need for the DG. Likewise, usage of the BB seems to be inversely correlated with the penalty cost.

6. Conclusions and future research

Stand-alone HRES are generally more suitable than single-source systems to supply power in off-grid applications, especially in remote areas with difficult access. However, these systems must be designed carefully and account for the stochasticity and uncertainty in supply data to serve the demand at high reliability and affordable cost. In this study, a stand-alone HRES that consists of PV panels and wind turbines, integrated with battery storage and diesel generator, is considered to implement a cost-minimization design approach. In order to capture uncertainty in supply power, a finite number of scenarios with uncertain probabilities are extracted from time-series data of both solar and wind power through k -means clustering. Next, to tackle the uncertainty in scenarios' probability, an ambiguity set based on the Variation Distance ϕ -divergence is constructed around the nominal probability of scenarios. Since TAC and reliability functions cannot be evaluated explicitly, two novel robust simulation-optimization methods, namely Classical RSM and GRST, are proposed to estimate the objective function from a surrogate model and optimize it. The presented methods are very generic and flexible as they can be used for different applications, and not only HRES design, with different objectives, operational rules, reliability metrics, scenario generation methods, ambiguity sets, etc.

The numerical experiments conducted on a realistic case study have shown the applicability and effectiveness of both methods. The optimal designs obtained using the classical RSM were found to marginally outperform the GRST designs in terms of TAC, whereas the reliability comparison results were mixed. However, the GRST is faster to

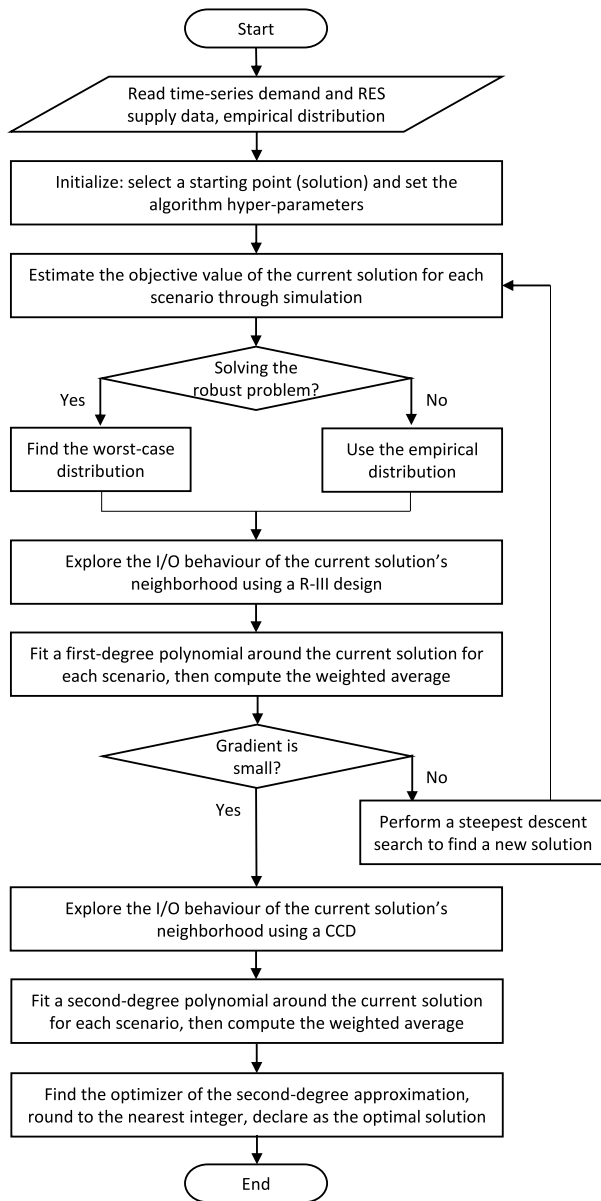


Fig. 10. Flowchart of the classical RSM.

converge and less dependent on the starting point than the classical RSM. Also, through cross-validation and hypothesis testing, it was shown that using the robust model with a properly-sized distributional ambiguity set leads to significantly better solutions than the nominal model. The effect of the penalty cost on TAC, reliability, system design and operation was also investigated and found to be quite significant. In particular, the Pareto frontiers constructed show a clear trade-off between TAC and reliability.

For future studies, other sources of renewable energy such as biomass, hydro, etc. can be used alongside/instead of PV/wind/battery/diesel units in order to build a robust and reliable HRES. Short-term (intra-hour) fluctuations in demand and RES supply can also be considered in future research, e.g., by using shorter time intervals and/or other energy storage technologies. Moreover, alternative metamodel-based approaches for simulation optimization such as Kriging and neural network can be implemented to estimate the objective function

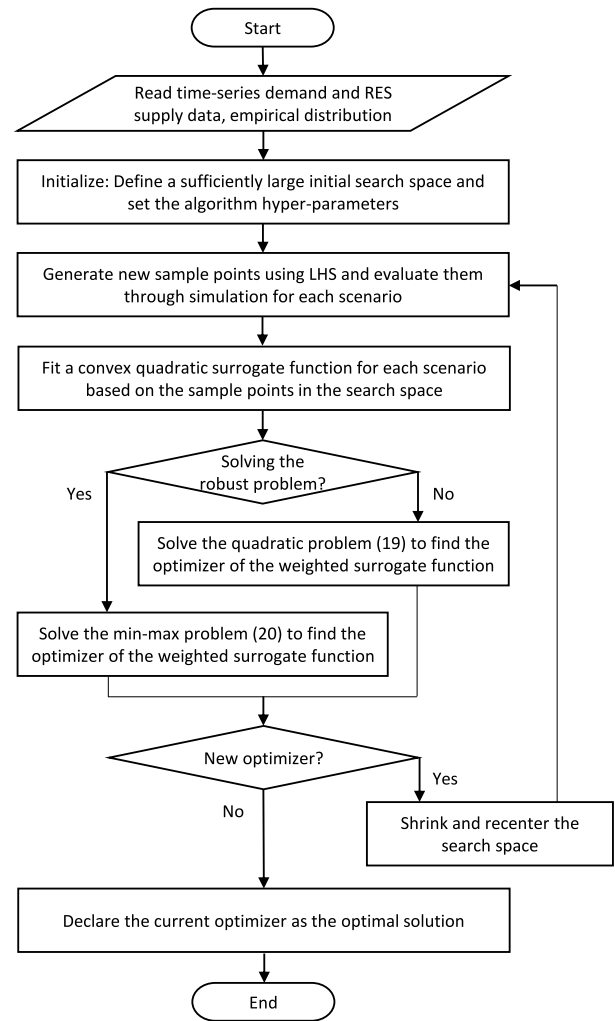


Fig. 11. Flowchart of the GRST.

as a surrogate model. Finally, other types of ϕ -divergence function such as Kullback–Leibler or χ^2 distance, can be utilized to construct the ambiguity set.

CRediT authorship contribution statement

Pardis Pourmohammadi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Ahmed Saif:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Ahmed Saif reports financial support was provided by Natural Sciences and Engineering Research Council of Canada.

Data availability

Data will be made available on request.

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Appendix. Flowcharts of the proposed methods

Flowchart representations of the classical RSM and the GRST are provided in Figs. 10 and 11, respectively.

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